Practical No.: 05

Code:

```
#include <p18f4550.h>
#include <delays.h>

#pragma config FOSC = HS
#pragma config WDT = OFF
#pragma config LVP = OFF
#pragma config PBADEN = OFF
#pragma config MCLRE = ON

void main(void)
{
  // RB0 as input (Switch)
  TRISBbits.TRISB0 = 1;

  // RD0 as output (LED)
  TRISDbits.TRISD0 = 0;

  // Make pins digital
  ADCON1 = 0x0F;
  CMCON = 0x07;

  // LED initially OFF
  LATDbits.LATD0 = 0;

  while(1)
  {
    if(PORTBbits.RB0 == 1)
    {
      // LED ON
      LATDbits.LATD0 = 1;

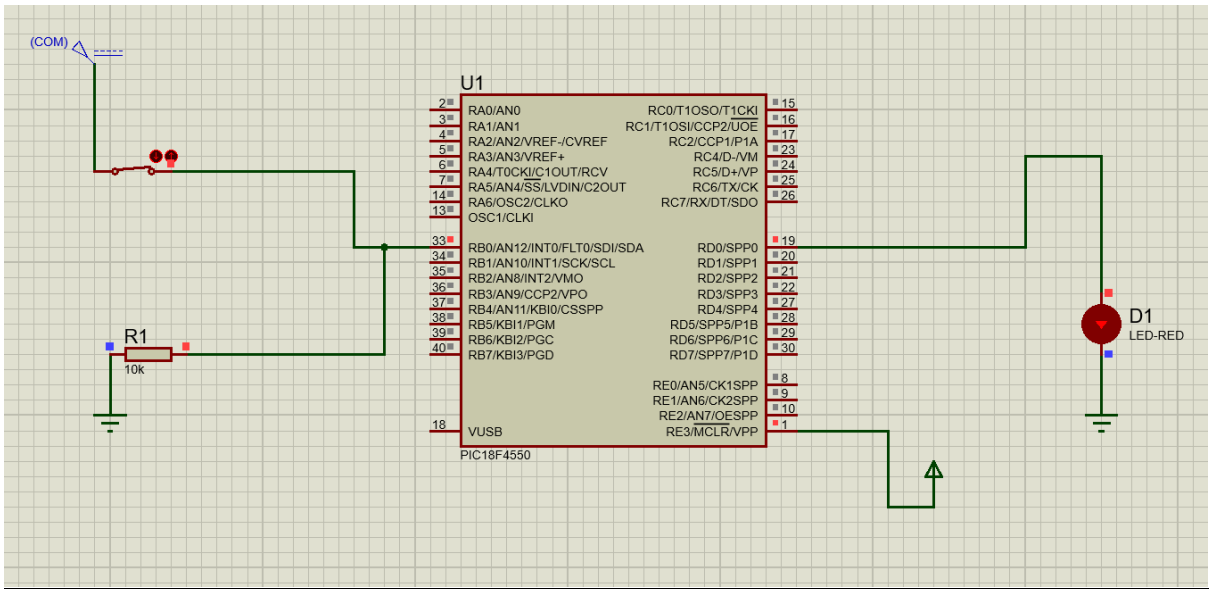
      // Wait for 2 seconds
      Delay10KTCYx(160);

      // LED OFF
      LATDbits.LATD0 = 0;

      // Wait for 2 seconds
      Delay10KTCYx(160);
    }
    else
    {
      // Switch OFF -> LED OFF
      LATDbits.LATD0 = 0;
```

```
    }
  }
}
```

Output:


(COM)
U1
RA0/AN0
RA1/AN1
RA2/AN2/VREF-/CVREF
RA3/AN3/VREF+
RA4/T0CKI/C1OUT/RCV
RA5/AN4/SS/LVDIN/C2OUT
RA6/OSC2/CLKO
OSC1/CLKI
RB0/AN12/INT0/FLT0/SDI/SDA
RB1/AN10/INT1/SCK/SCL
RB2/AN8/INT2/VMO
RB3/AN9/CCP2/VPO
RB4/AN11/KBI0/CSSPP
RB5/KBI1/PGM
RB6/KBI2/PGC
RB7/KBI3/PGD
VUSB
RC0/T1OSO/T1CKI
RC1/T1OSI/CCP2/UOE
RC2/CCP1/P1A
RC4/D-/VM
RC5/D+/VP
RC6/TX/CK
RC7/RX/DT/SDO
RD0/SPP0
RD1/SPP1
RD2/SPP2
RD3/SPP3
RD4/SPP4
RD5/SPP5/P1B
RD6/SPP6/P1C
RD7/SPP7/P1D
RE0/AN5/CK1SPP
RE1/AN6/CK2SPP
RE2/AN7/OESPP
RE3/MCLR/VPP
PIC18F4550
R1
10k
D1
LED-RED

Practical No. 06

Code:

```
#include <p18f4550.h>
#include <delays.h>

// Configuration Bits
#pragma config FOSC = HS      // High-Speed Oscillator
#pragma config WDT = OFF      // Watchdog Timer Disabled
#pragma config LVP = OFF      // Low-Voltage Programming Disabled
#pragma config PBADEN = OFF    // PORTB<4:0> configured as digital I/O
#pragma config MCLRE = ON     // MCLR pin enabled

// Pin Definitions matching your schematic
#define IN1   LATBbits.LATB0    // RB0 -> L293D Pin 2 (IN1)
#define IN2   LATBbits.LATB1    // RB1 -> L293D Pin 7 (IN2)
#define EN1   LATBbits.LATB2    // RB2 -> L293D Pin 1 (EN1)

// Function for software PWM speed control on EN1
// duty_cycle: 0 to 100 | duration_cycles: controls how long it runs
void motor_pwm(unsigned char duty_cycle, unsigned int duration_cycles)
{
  unsigned int i;
  for(i = 0; i < duration_cycles; i++)
  {
    if(duty_cycle > 0)
    {
      EN1 = 1;
      // Delay for ON time proportional to duty cycle
      Delay10TCYx(duty_cycle * 2);
    }
    if(duty_cycle < 100)
    {
      EN1 = 0;
      // Delay for OFF time proportional to (100 - duty cycle)
      Delay10TCYx((100 - duty_cycle) * 2);
    }
  }
}

void delay_1sec(void)
{
  // Generates approx 1 second delay
  Delay10KTCYx(100);
  Delay10KTCYx(100);
}
```

```
void main(void)
{
  // Set RB0, RB1, RB2 as outputs
  TRISBbits.TRISB0 = 0;
  TRISBbits.TRISB1 = 0;
  TRISBbits.TRISB2 = 0;

  // Set digital I/O mode
  ADCON1 = 0x0F;
  CMCON  = 0x07;

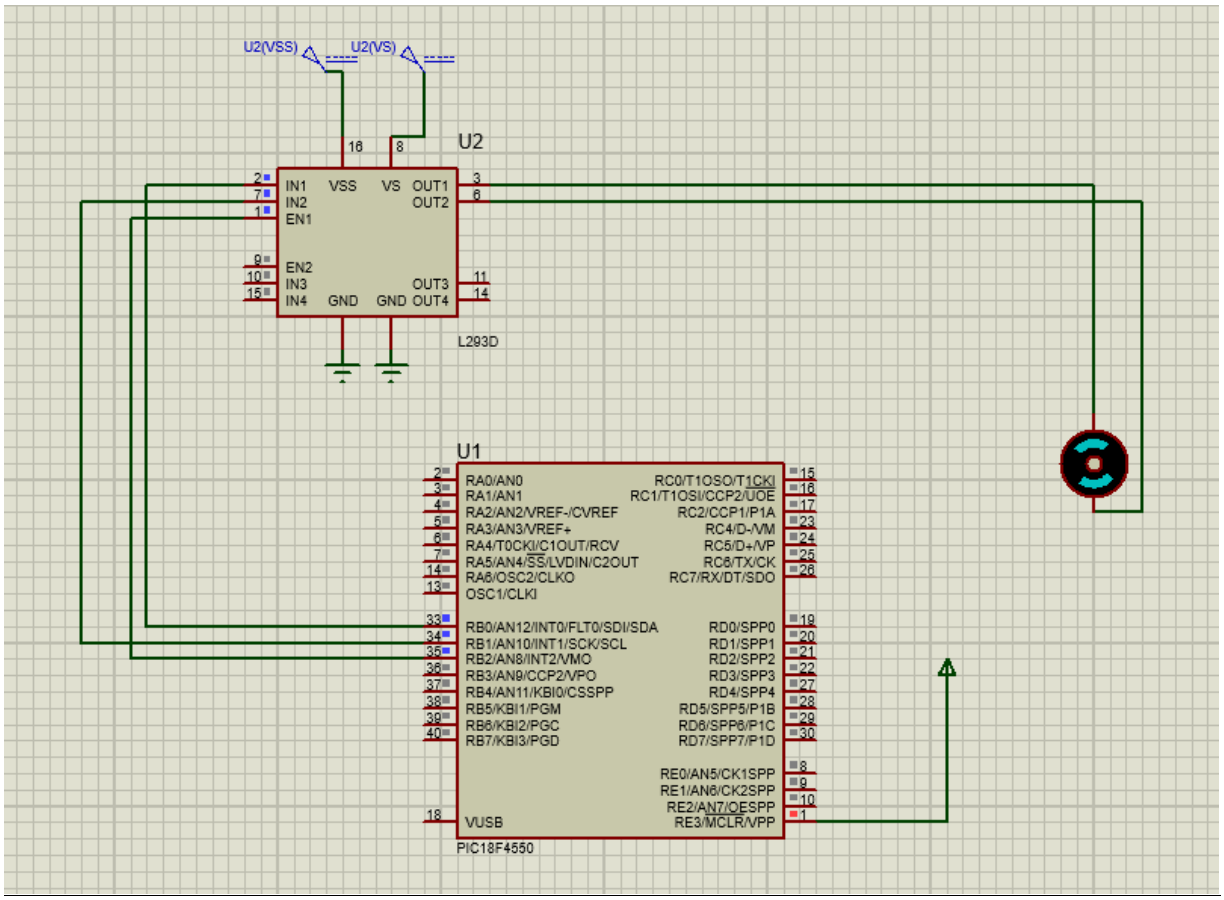
  // Initialize motor to STOP
  IN1 = 0;
  IN2 = 0;
  EN1 = 0;

  while(1)
  {
    // 1. Clockwise Rotation @ 70% Speed (Medium)
    IN1 = 1;
    IN2 = 0;
    motor_pwm(70, 600);    // Runs clockwise

    // Stop Motor for 1 second
    IN1 = 0;
    IN2 = 0;
    EN1 = 0;
    delay_1sec();

    // 2. Anticlockwise Rotation @ 90% Speed (High)
    IN1 = 0;
    IN2 = 1;
    motor_pwm(90, 600);    // Runs anticlockwise

    // Stop Motor for 1 second
    IN1 = 0;
    IN2 = 0;
    EN1 = 0;
    delay_1sec();
  }
}
```


U2(VSS)
U2(VS)
16
8
U2
IN1
IN2
EN1
VSS
VS
OUT1
OUT2
EN2
IN3
IN4
GND
GND
OUT3
OUT4
L293D
U1
RA0/AN0
RA1/AN1
RA2/AN2/VREF-/CVREF
RA3/AN3/VREF+
RA4/T0CKI/C1OUT/RCV
RA5/AN4/SS/LVDIN/C2OUT
RA6/OSC2/CLKO
OSC1/CLKI
RB0/AN12/INT0/FLT0/SDI/SDA
RB1/AN10/INT1/SCK/SCL
RB2/AN8/INT2/VMO
RB3/AN9/CCP2/VPO
RB4/AN11/KBI0/CSSPP
RB5/KBI1/PGM
RB6/KBI2/PGC
RB7/KBI3/PGD
VUSB
RC0/T1OSO/T1CKI
RC1/T1OSI/CCP2/UOE
RC2/CCP1/P1A
RC4/D-/VM
RC5/D+/VP
RC6/TX/CK
RC7/RX/DT/SDO
RD0/SPP0
RD1/SPP1
RD2/SPP2
RD3/SPP3
RD4/SPP4
RD5/SPP5/P1B
RD6/SPP6/P1C
RD7/SPP7/P1D
RE0/AN5/CK1SPP
RE1/AN6/CK2SPP
RE2/AN7/OESPP
RE3/MCLR/VPP
PIC18F4550